A Biochemical Study of the Growth of *Drosophila melanogaster*

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ABSTRACT 1. The growth of a wild strain of *Drosophila melanogaster* has been studied from the time of hatching from the egg to the adult. Wet and dry weights, protein and protein fractions, lipids, RNA and DNA have been determined at successive intervals. 2. All estimates have been carried out on animals grown axenically on defined medium. The medium has been shown to permit growth which is as fast as on live yeast medium when the larvae can burrow freely. 3. Insoluble protein and chitin nitrogen reach a maximum of about 70% total nitrogen in the late pupa, at which time the soluble nitrogen reaches its lowest level. The amino acid fraction reaches a maximum in the early third instar. 4. The water content declines from 70–75% of the live weight at the end of the first instar, rises to a maximum during the second instar and declines sharply to about 66% at the prepupal stage. 5. Total lipids represent an increasing proportion of body weight during larval growth, increasing from about 6% in young larvae to about 15% at pupation. 6. The curve of RNA increase follows the dry weight and protein curves during larval life. The ratio of RNA to protein declines during the first two instars. 7. The increase in DNA content follows a path similar to that of RNA. The RNA/DNA ratio reaches a maximum 24 hours after hatching from the egg, declines to about half this value by 84 hours, and remains more or less constant thereafter. 8. The rates of increase of protein, RNA and DNA during successive periods of larval life show striking differences. For protein the rate is high and increasing until about 36 hours after which it declines rapidly during the rest of larval life. For both nucleic acids the rate of increase falls to a minimum in the early third instar, about the time of the critical size, then increases during proliferation of the imaginal discs and finally declines at the end of larval life. 8. The data are compared with evidence from other strains of *Drosophila* and from other species of insect. In the latter case, it appears that all the *Diptera*, so far examined, follow a similar trend in the increase of the nucleic acids during growth. 9. RNA synthesis has been followed during development from egg to adult with the aid of pulse-labeling and sucrose gradient analysis. Developmental stages differ in the amount of rapidly labeled high molecular weight material which is most evident in newly hatched larvae and in mid-third instar larvae, at times when the rate of synthesis of total RNA is high.

Studies on the growth of *Drosophila* have been concerned either with records of wet and dry weight (Bakker, '59), or the DNA content (Nigon and Daillie, '58) of whole animals or with comparisons between genotypes, especially in certain tissues, such as the salivary glands (Patterson and Dackerman, '59; Altorfer, '53; Chen, Farinella-Ferruzza and Delhafen-Gandolla, '63) or fat body cells (Welch, '57). A more complete biochemical description of growth is needed for detailed comparisons between genotypes within the species and also for comparisons with insects from other families. Various parameters of growth such as protein, DNA etc. have been established for *Aedes* (Lang, Lau and Jefferson, '65), *Calliphora* (Price, '65), *Tribolium* (Devi, Lamonde, Srivasata and Laker, '63) and *Tenebrio* (Teissier, '31) but information for a greater range of insects would be valuable.

The principal reason for the present investigation was to provide a basis for comparing differences in growth between lines of *D. melanogaster* cultured axenically. These lines had been selected for large or small body size or longer or shorter development time on various chemically defined axenic media (Robertson, '60, '64) and displayed striking differences in growth rate and the duration of development, suggesting the need for biochemical comparison (Church and Robertson, in press).

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The present paper describes the growth of a wild, unselected, strain of *D. melanogaster* on optimal axenic media in terms of wet and dry weight, protein, lipid, DNA and RNA content. Characterization of *Drosophila* nucleic acids at successive stages of development will be presented elsewhere.

MATERIAL

The Pacific strain of *D. melanogaster* used in this investigation was derived from a number of flies caught on the Pacific coast of the United States, and has been maintained as a large population for several years. The flies used for the various estimations have been cultured on the usual live yeast cornmeal molasses medium or reared under axenic conditions on a chemically defined medium, which was essentially the same as medium C of Sang ('56). The nutrients, which are dissolved or suspended evenly in an agar gel, include casein, yeast nucleic acid, lecithin, fructose, cholesterol, calcium pantothenate, nicotinic acid, thiamine, pyridoxin, biotin and folic acid, together with a buffer solution. Autoclaving time and pressure were kept to a minimum to avoid inactivation of amino-acids (Evans and Butts, '49) or fructose oxidation (Sang, '56). Eggs were dechorionated in hypochlorite, sterilized and introduced aseptically into cultures by the procedure outlined by Robertson ('60). Larvae were washed with sucrose and insect Ringer after heating the culture to drive the animals to the surface.

The measurement of growth

(a) *General.* Usually growth is expressed in terms of dry or wet weight or in linear dimensions. Perhaps the best approach to a general method would be to describe the chemical development of tissues in terms of the changes in cell number and in the amounts of the various constituents per cell. Davidson and Leslie ('50) advocated the use of DNA per nucleus and the total DNA per animal to provide an estimate of cell number. The constancy of DNA per nucleus, where applicable, bypasses the complications caused by allometric relations since it provides an unambiguous index. Also, the relative importance of changes in cell size and number can be estimated for tissues which differ in mass. In insects, however, in which growth may be associated with a great increase in cell size alone, accompanied by high levels of polyteny in certain tissues, this method must be applied with caution, although the estimation of total DNA per individual, at successive stages of growth, has much to commend it. In the present material the amounts of several constituents were followed during growth, in addition to wet and dry weights and so it is possible to make instructive comparisons between their ratios at different stages.

(b) *Weight.* Individual larvae, pupae, adults and batches of ten eggs were weighed on a Cahn Electrobalance. Adult weight was determined either by subtracting the weight of the puparium from that of the late pupa or by weighing the newly emerged adult. The former method is preferable (Royes, '63). Dry weights were obtained after three hours drying at 100°C.

(c) *Protein.* Several different methods of estimating protein were compared on replicated samples of eggs, late larvae, early pupae or adults. These were as follows:

(i) 1–2 mg tissue was homogenized in 10 × 150 mm pyrex micro-Kjehldahl tubes in 2 ml H_2SO_4 containing 0.02 ml catalyst ($CuSo_4$, H_2SO_4, H_2O, Se in the ratio 5:15:5:1) plus one drop of 30% H_2O_2. After overnight digestion at 100°C the distillate was collected in 2 ml 4% boric acid and titrated with 0.01N HCl, according to Markham ('42). The colorometric procedure of Umbreit, Burris, and Stauffer ('64) was also used.

(ii) 0.5–5 mg tissue was homogenized in 2 ml 2% K acetate in 90% ethanol, to extract lipids and prevent proteins going into solution (Steele, Okomura and Busch, '64), and the sample was centrifuged at 15,000 rpm for five minutes. After removal of the supernant, 1 ml of water was added. The mixture was buzzed, 2 ml of 5% TCA were added; the preparation was buzzed and centrifuged as before. A further 1 ml of water was added to the residue and, after buzzing, the sample was dialyzed against 0.01 M triethanolamine to remove the uric acid and free

nucleotides. The technique of Lowry, Rosebrough, Farr and Randall ('51) was followed thereafter. Absorption was read against a blank at 500 and 750 mμ. Standard curves were prepared from bovine serum albumin.

Drosophila excretes excess nitrogen as uric acid, which, if not allowed for, can lead to spuriously high estimates of protein nitrogen. Uric acid, estimated by the procedure of Bernard and Fixler ('63), was found to be approximately 1 μg per animal for larvae and adults and 5 μg for pupae. All nitrogen estimations were corrected for uric acid before estimating the protein content as 6.25 × N.

The effects of dialyzing the tissue homogenates were determined for the alternative procedures. In the micro-Kjehldahl method, the differences in estimated nitrogen were negligible, but, in the modified Lowry procedure, dialysis led to lower estimates, especially in pupae, and indicated removal of uric acid. Since there was excellent agreement between the Kjehldahl and the Folin-Ciocalteau estimates on dialyzed tissue, and also excellent repeatability with the latter method, it was the obvious choice for routine protein determination. In later stages of development the total nitrogen was fractionated by the procedure of Anderson ('48).

(d) *Lipid estimation.* Estimations of lipid content were carried out on 0.1 g material; this involved variable numbers of individuals, depending on their age. The procedure of Folch, Lees and Stanley ('57) was used. The tissue was homogenized extensively with a 20:1 (v/w) mixture of chloroform-methanol (2:1) and after three hours, the extract was washed with 0.3 volume of 0.05 N NaCl. The mixture was centrifuged to separate the two phases, the lower of which contained the total pure lipid extract. This was filtered through fat-free filter paper, and, after drying, the lipid content was determined from the difference in weight. The 95% alcohol-ether extraction procedure of Finkel ('48) failed to give repeatable estimates with the *Drosophila* material.

(e) *The estimation of DNA and RNA.* The accurate estimation of the nucleic acid content of tissues poses many problems which have been recently discussed by Hutcheson and Munro ('61). Each kind of material may require special modifications of procedure which have to be determined empirically. Many procedures were tested on the *Drosophila* material before adopting the methods described below. Levenbook, Travaglini and Schultz ('58) showed that estimates may be inaccurate if there are differences in the base ratio between the DNA used as a standard and that extracted from the tissues, so all estimates of DNA and RNA content are based on nucleic acids extracted from *Drosophila* (Church, in press).

The number of animals used for nucleic acid estimations varied from 50 first instar to three late third instar larvae, five pupae and three adults per sample. Duplicate estimations were always carried out. Larvae were removed from the medium at the appropriate ages, washed in insect Ringer solution, and dried on filter paper.

The animals were then transferred to an ice cold 10 × 60 mm silicone treated pyrex centrifuge tube. Homogenization was carried out in 0.5 ml cold 95% ethanol (v/w) with a siliconed glass rod shaped to fit the conical tip of the tube. After 15 minutes the homogenate was centrifuged at 2,000 g for five minutes and the supernatant pipetted off. 0.5 ml methanol-chloroform mixture (2:1, v/w) was added and the sample was buzzed before being left for 15 minutes at 4°C and then centrifuged for ten minutes at 2,000 g. If the supernatant was still cloudy, as was generally the case with third instar larvae, another methanol-chloroform extraction was carried out.

To the lipid-free residue was added 0.5 ml ice cold 0.3N PCA and the mixture was homogenized and buzzed briefly. After ten minutes at 0°C an additional 0.15 ml of 0.3N PCA was added and the mixture was buzzed and centrifuged as before for ten minutes. Tests showed that acid soluble phosphorus extraction was virtually complete with this procedure. For third instar larvae and pupae, the PCA mixture was dialyzed against 0.01 M triethanolamine (pH 7.0) at 45°C for two hours to remove the uric acid before centrifugation.

Two methods of extracting nucleic acids from the lipid-free residue were used, namely the procedure of Schneider, Hogeboom and Ross ('50), and the Fleck and Munro ('61) modification of the Schmidt and Thannhauser ('45) method. The Ogur and Rosen ('50) procedure led to varying amounts of DNA in the RNA fraction.

In the Schneider, Hogeboom and Ross ('50) procedure, two extractions were carried out at 80°C for 20 minutes with 0.5 ml of 1N PCA.

In the Schmidt and Thannhauser method, the residue was hydrolyzed at 37°C for two hours in 0.5 ml 0.3 N KOH. The combined supernatants were acidified (pH 2.0) with cold concentrated PCA to precipitate DNA, protein and $KClO_4$. After spinning, the residue was washed with 0.25 ml 0.1N $KClO_4$ and centrifuged as before. RNA determinations were carried out on the combined supernatants, while the residue was extracted twice with 0.5 ml 1N PCA at 80°C for 15 minutes and the DNA was determined on the combined extracts.

For the estimation of RNA the orcinol reaction of Mejbaum ('39), as described by Schneider ('57), was used. The Dische ('30) reaction, as modified by Burton ('56), was used for DNA. Both the diphenylamine and orcinol were purified by recrystallization from hot hexane.

Preliminary experiments showed that the same values of RNA were obtained either by use of the orcinol reaction or by the difference between diphenylamine estimates of DNA and the estimates for the total nucleic acids, derived by U.V. absorption at 260 mμ of the extract prepared by the method of Schneider, Hogeboom and Ross ('50).

For the estimation of the DNA content in eggs and very young larvae, for which only limited amounts of material of precisely determined age were available, an enrichment technique utilizing 1.0 ml of a DNA solution was prepared by dissolving 50 μg DNA per ml in 0.5 N PCA at 90°C. DNA was estimated by difference between PCA blanks and a standard curve. Excellent proportionality between optical density and DNA content, estimated in this way, was noted although the standard error was rather high.

(f) *Repeatability of estimates.* Table 1 shows the repeatability of the various estimations expressed as the coefficient of variation. The first column of coefficients refers to estimates carried out on animals up to the end of the first instar, while the second column refers to the remainder of development. The repeatability for protein determination is very high; coefficients range from 1–2.4%. The repeatability of DNA and RNA is satisfactory, especially for U.V. determinations.

TABLE 1

The repeatability of biochemical determinations

Constituent estimated	Coefficient of variation (%)		Relative repeatability
	Egg/ 1st instar	2nd instar — Adult	
Protein			
Lowry	1.1	2.4	high
N estimates	1.3	2.1	high
Lipid	11.0	20.0	low
RNA			
Orcinol	6.4	3.4	medium high
UV	—	2.6	high
DNA			
Diphenylamine	8.5	5.2	medium
UV	—	2.1	high
Total phosphorus	2.0	3.1	high
Total nucleic acids	2.1	1.4	high

(g) *RNA synthesis during development.* A preliminary study of RNA synthesis has been carried out with the aid of labeling with P^{32} and sucrose gradient centrifugation. For qualitative analysis of the RNA synthesis at successive stages, a modification of Brown and Littna's ('64) method was used. Axenically cultured larvae were homogenized in 0.5% sodium lauryl sulphate, 0.1 M NaCl, 0.1 M Na acetate buffer (pH 5.5) containing 5 mg bentonite at 4°C. The suspension was shaken for ten minutes at 4°C after adding an equal volume of phenol (pH 5.5) containing 0.1% 8-hydroxyquinoline. The aqueous supernatant, after centrifugation, was removed and the RNA precipitated by addition of 0.1 volume of Na acetate and two volumes of cold ethanol. To remove DNA, two μg DNase per ml was added in 0.001 M $MgCl_2$ and incubated for ten minutes at 10°C. After centrifuging, the pellet was taken up in acetate buffered 5% sucrose containing 1 mg bentonite per ml and 0.5% sodium lauryl sulphate. 1 ml of this was back-layered onto a 5–20% buffered sucrose gradient and spun for 18 hours at 2,300 rpm at 20°C in a S25.2 rotor in a model L2 Spinco ultracentrifuge.

For pulse-labeling tests, washed larvae were placed in 0.5% sucrose to which 200 μc of neutralized P^{32} orthophosphate had been added. Two hours later the larvae were removed, cooled below 4°C and stored in sucrose-buffered solution until extraction of the RNA.

The sedimentation analysis was carried out at successive stages of larval life. *Drosophila* ribosomal RNA of known size was used as a marker and located by its optical density at 260 mμ. The scale, for measurement of radioactivity, varies for the successive tests and so only qualitative differences in the profiles can be considered. Rapid extraction and stringent precaution against RNase activity has been the rule for all tests.

RESULTS

Growth records

(1) *Weight changes.* The wet weight growth curves for females of the Pacific stock, grown either on the live yeast or the axenic medium, are presented in figure 1. Also shown are similar data,

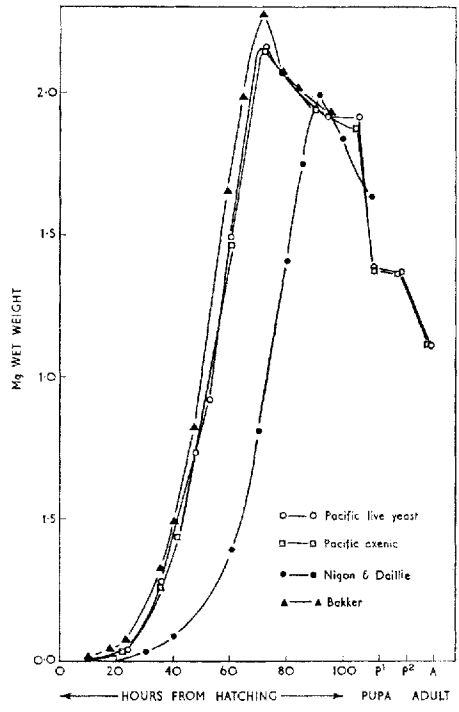

Fig. 1 The growth of three different strains on live yeast medium and also of the Pacific strain on the axenic diet. P^1 and P^2 refer respectively to estimates made within five hours of pupation (eversion of the spiracles) and within 20 hours of eclosion.

for different strains, reported by Bakker ('59), and Nigon and Daillie ('58). On the live yeast medium, the Pacific strain reached a maximum wet weight of 2.162 mg at 72 hours after hatching from the egg. Growth rate was fastest until 48 hours and slowed down thereafter. Bakker's strain reached a slightly greater wet weight of 2.280 mg at 72 hours, but this strain apparently develops faster since pupation occurred at 90 hours, compared with 104 hours in the Pacific strain. In the growth curves reported by Nigon and Daillie ('58), a lower maximum weight of 1.980 mg was reached at a later time, 90 hours, and pupation occurred at 112 hours. This slower growth may be due to less favorable nutrition and/or genetic differences. Inbreeding, for example, would favor a change of this kind.

Tests have shown that when larvae are grown on the axenic medium there is an initial delay due to the skin on the gel

surface which prevents newly hatched larvae from feeding to their capacity until they have reached a certain size and can burrow freely. On the live medium there is no such problem since the larvae can feed on the yeast growing on the surface. If the surface of the axenic medium is scarified this delay can be reduced to about 24 hours. When the growth curves for the axenic diet are corrected to allow for this initial delay they virtually coincide with the curves for growth on the live yeast medium (fig. 1), thereby validating the use of this correction and also demonstrating the adequacy of the axenic diet when the larvae can exploit it fully.

The decline in wet weight of the last stage of larval life coincides with the time when the larvae leave the medium and crawl to a pupation site. No loss of weight occurs during the pupal period, presumably due to the impermeability of the cuticle (Ludwig, '45).

The curves for dry weight for the Pacific and Bakker's strain grown on live yeast are shown in figure 2. There is no significant difference in dry weight between Pacific stock reared on live yeast or axenic optimal medium. The relative differences in size indicated for the wet weight also hold for dry weight, which attains a maximum of 0.487 mg and 0.537 mg respectively in the Pacific and Bakker strains at 84 hours, i.e., ten hours later than the wet weight maximum. The decline in dry weight thereafter is associated with catabolism of lipid reserves and histolysis of larval protein. It will be shown later that there is a 10% decline in lipid content between prepupa and adult. In both strains the dry weight of adult females is about 70% the maximum dry weight of the larvae. More than 80% of this difference in dry weight is accounted for by the puparium. From about 48 hours onwards the sexes can be distinguished by gonad size. In both strains the dry weight of males is about 75% that of females at corresponding stages.

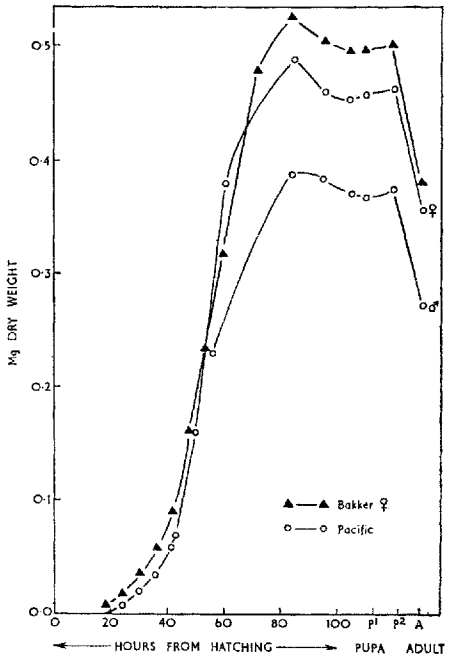

Fig. 2 Comparison of growth in terms of dry weight of the Pacific and Bakker's strain. Symbols as in figure 1. Animals were grown on live yeast medium.

(2) *Protein.* The protein content of the Pacific strain, cultured on the axenic medium, is shown in figure 3, which also includes estimates of the distribution of protein nitrogen from the second instar to the adult. The nitrogen was partitioned into three fractions: I, the complex water insoluble protein and chitin; II, the soluble protein and peptide nitrogen, and III, the free amino acid nitrogen, after removal of uric acid by dialysis. The changes in these fractions are expressed as percentages of the total nitrogen.

The total protein content is very closely correlated with wet and dry weight until after 72 hours of development, when the protein synthesis in the proliferating imaginal discs increasingly influences the total protein content of the animal. A maximum of 0.140 mg was reached in the late pupa in females. This represents the joint contribution from larval protein, which has not undergone histolysis, and the adult protein.

With respect to the nitrogen distribution, Fraction I, containing insoluble protein and chitin nitrogen, reached a maximum of 70% in the late pupa, at which time the soluble protein nitrogen of Fraction II reaches its lowest level, suggesting that, by this time, the construction of the adult exoskeleton had been completed.

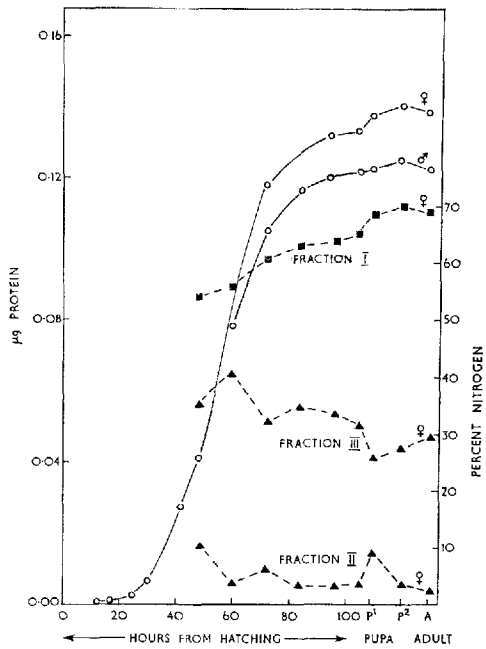

Fig. 3 Changes during growth of protein content and also the distribution of the protein nitrogen into three fractions (Anderson, '48). Fraction I represents insoluble protein and chitin nitrogen. Fraction II represents soluble protein nitrogen and Fraction III the free amino-acid level. Symbols as in figure 1. Animals were grown on optimal axenic medium.

Pupation was accompanied by relatively large changes in the distribution of nitrogen. The amino acid level, Fraction III, showed a decline, while the soluble protein fraction increased. This interrelationship probably reflects the rapid growth of structural proteins, the developing imaginal discs, at the expense of the free amino-acid pool. The leveling off of the rate of change in these two fractions, as pupal development proceeds, may be related to two indistinguishable processes, i.e., histolysis of larval and synthesis of adult protein.

The amino-acid fraction (III) presumably represents the compounds available for protein synthesis. This fraction is higher in the early third instar larvae than during any later stage of development and represents 40% of the total nitrogen at that time. This level is consistent with the high levels of hemolymph amino acids and of soluble peptides found in most insects (Wigglesworth, '54).

(3) *Percentage composition of wet weight.* Figure 4 shows the changes in lipids, protein and water during development, on axenic optimal medium, expressed as a percentage of live weight. As might be expected, water accounts for the greater part of the total weight. It declines from 70 to 75% at the end of the first instar, rises to a maximum of 83% during the second instar, and then declines sharply to 66% in the prepupal stage. As noted earlier, this corresponds to the time when the larva leaves the medium.

The percentage protein remains fairly constant at 4–6% until about 72 hours when it begins to increase steadily to reach 11.5% in the adult. Two important peaks can be discerned during the course of growth, at 18 and 42 hours, both of which correspond to pre-moult periods when the larva is synthesizing new integument. The late larval and pupal increase coincides with the development of imaginal organs and the thoracic musculature.

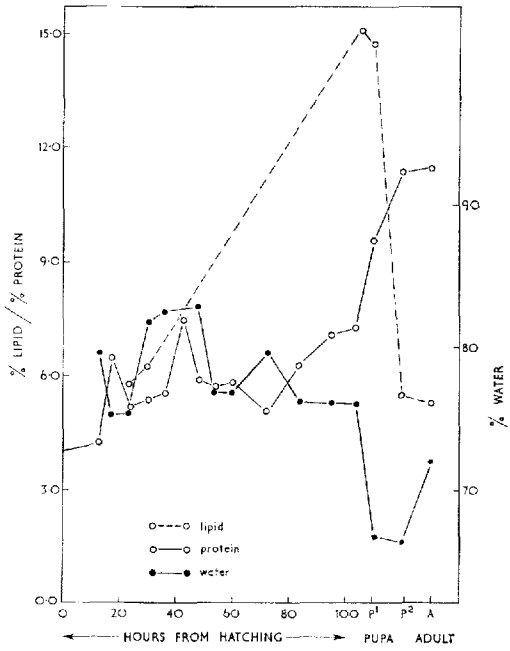

Fig. 4 Composition in terms of per cent lipid, water and protein during growth. Symbols as in figure 1. Animals were grown on optimal axenic medium.

Definite trends are apparent in the lipid content during growth. Total lipids represent an increasing proportion of body weight, ranging from 6% at 24 hours to 15% at puparium formation. During the pupal period the lipid content declined by 10%. As in other holometabolous insects (Wigglesworth, '54), there is a large larval fat body which increases during larval life. The increase in lipids may be due to increase in neutral fat content, as reported for the mealworm by Finkel ('48).

(4) Nucleic acids. Figure 5 shows the RNA content per animal cultured on axenic optimal media for the two sexes and also the ratio of RNA (μg) to protein (mg). The RNA curve generally follows the dry weight curve up to 84 hours. The deviations from a smooth curve, before the sexes can be distinguished, are probably related to differences between the sexes in moulting time and general variation within the sexes in this respect. The ratio of RNA to protein, illustrated for females only, shows a decrease in RNA per unit protein during the first two instars, although this is preceded by a marked synthesis of RNA during embryogenesis, which continues until about 12 hours after hatching.

Figure 6 shows the comparable data for DNA for animals cultured on axenic optimal media. The DNA content follows the protein curve during development. The curve of the ratio to protein is very similar to the corresponding curve for RNA. In both a peak occurs at about 12 hours after hatching from the egg although the RNA is a little ahead of the DNA peak. It is followed by a rapid decline to a minimum at about 60 hours of 35 μg RNA and 3.5 μg DNA per mg protein. This represents respectively a 70 and a 40-fold reduction below the maximum values. After 60 hours the ratios gradually increase to reach levels in the adult of 50 μg RNA and 7.4 μg DNA per mg protein.

It is also instructive to consider growth in terms of the ratio of either RNA or protein to DNA (fig. 7). A maximum

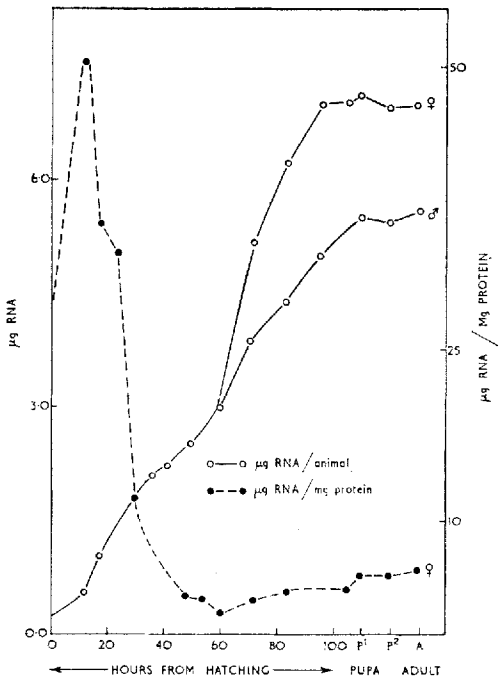

Fig. 5 RNA content per individual during growth and also the ratio of RNA (microgram) to protein (milligram). Symbols as in figure 1. Animals were grown on optimal axenic medium.

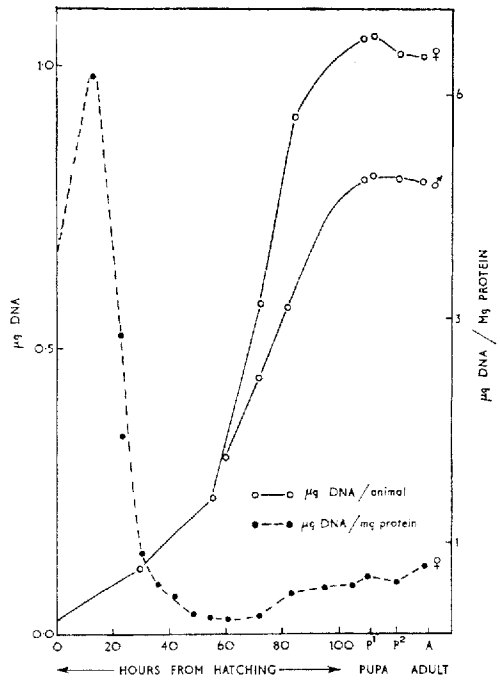

Fig. 6 DNA content per individual during growth and also the ratio of DNA (microgram) to protein (milligram). Symbols as in figure 1. Animals were grown on otpimal axenic medium.

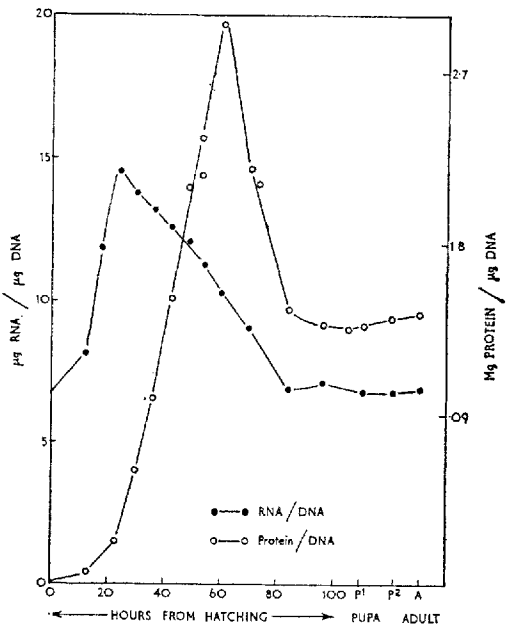

Fig. 7 The ratio of RNA (microgram) to DNA (microgram) and also protein (milligram) to DNA (microgram) during growth of animals grown on the otpimum axenic diet. Symbols as in figure 1.

RNA/DNA ratio of 14.5 is reached 24 hours after hatching from the egg, after which the value declines to 8.9 at 84 hours and remains at this level thereafter. For protein the maximum of 0.296 mg per μg DNA is reached at 60 hours, after which there is a sharp decline to 0.136 mg per μg DNA at about 84 hours followed by constancy at this level. These relations imply that rapid DNA synthesis during larval development, is accompanied by a rapid synthesis of RNA which is, in turn, succeeded by a maximum ratio of protein to DNA, suggesting that growth by cell size predominates. The leveling off of the ratios after 84 hours is associated with the proliferation of imaginal cells. Nigon and Daillie ('58) reported that the DNA content rose sharply after the larval weight reached 1.3 mg and Neulat ('61) attributed this to the growth of the imaginal discs. There is an apparent discrepancy between these workers' results and those described in figure 6. It is perhaps relevant that Combe, Penot, Charlier and Sacquet ('65) found that the DNA content of the intestine of "germ-free" rats was 20% below that of normal controls due, presumably, to the absence of symbionts. It is possible that enhanced DNA content of non-axenic *Drosophila* is also due to symbionts in the gut and perhaps other tissues as well, and, if so, this would partly account for the differences in DNA content reported for the two *Drosophila* strains. But it does not adequately dispose of the sharply increased DNA synthesis reported by Nigon and Daillie ('58), and it is probable that differences in the method of estimating DNA content are chiefly responsible.

The rate of increase of protein, RNA and DNA

It is particularly instructive to consider growth in terms of the rate of change of the principal constituents during larval life. For this purpose the content per individual of protein, RNA and DNA has been converted to natural logarithms. The difference between the log values in successive 12 hour periods conveniently summarizes the sequence of changes in rate (fig. 8). The initial rate of increase for

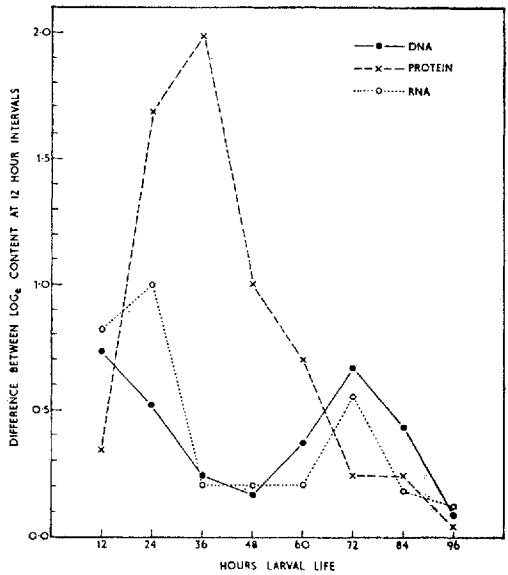

Fig. 8 The changes in the rate of increase of DNA, RNA and protein during larval life, expressed as the differences in the natural logarithm of the amount of each constituent per individual, at successive 12 hour periods.

protein is very high until about 36 hours of larval life, after which it declines rapidly and continuously throughout the rest of larval life. For DNA, after hatching from the egg, the rate of increase falls steadily until about 48 hours, after which it increases to about 72 hours and then declines. The increase in late larval life marks the increase in cell division associated with growth of the imaginal discs. In early larval life, the contrast between the positive increase in rate of increase for protein and the negative rate for DNA reflects the growth of larval tissues by increase in cell size. The curve for RNA follows the DNA pattern, except that the rate increases up to 24 hours after hatching from the egg.

The critical size, i.e., the time at which larvae can complete development even if no longer allowed to feed (Bakker, '59; Robertson, '63), occurs at 48 hours, i.e., at the time when the rate of increase of both DNA and RNA is minimal, immediately before the increased rate associated with proliferation of the imaginal discs. It is significant that the critical size, which probably coincides with an important shift in hormonal balance, occurs between periods of declining and increasing rates of increase of both nucleic acids.

Qualitative differences in RNA during development

Figure 9a-b shows the profiles whose features may be summarized as follows:

(a) shows the typical profile for RNA from newly laid eggs. 0.011 ml of C^{14} labeled mammalian ribosomal RNA was added for reference points at 28 S and 18 S. The optical density refers to the *Drosophila* RNA which shows 28 S and 18 S peaks, a soluble 4 S peak and also a small 30 S fraction which orcinol and diphenylamine reactions showed to be made up chiefly of DNA. Assuming that degradation had not occurred, about 90% of the RNA is ribosomal.

(b) shows the result of P^{32} labeling of newly hatched larvae. There is evidence of rapidly labeled high molecular weight material which probably represents DNA-like RNA and precursors of rRNA. The 4 S region in this and later profiles is occupied by a mixture of compounds, mostly degradation products resulting from the DNAase treatment used in the preparation of RNA. These compounds were not removed since attempts to do so resulted in complete degradation of the RNA.

(c) shows the profile for 12 hour larvae. Here there is evidence of synthesis of 28 S and 18 S RNA in the two labeled peaks. The relative amount of labeled, high molecular weight material has decreased.

(d) shows the profile at 48 hours. The label is confined to 28 S and 18 S material and there is no evidence of the high molecular weight material.

(e) shows the profile at 65 hours. There is again evidence of high molecular weight material and there is also relatively more label in the 28 S and 18 S components. At this time, when RNA is being synthesized at an accelerated rate (fig. 5) it would not be surprising to find increased evidence of messenger RNA and the high molecular weight material in the profile may be evidence of this.

(f) shows the profile at 84 hours. P^{32} incorporation is very low and there is only slight indication of 28 S and 18 S labeling.

DISCUSSION

The data presented above refer to a particular strain of *Drosophila melanogaster*. Wild strains are known to differ in body size and growth rate (Robertson, '59), and hence minor differences in the growth parameters may be expected, as in differences between the Pacific strain and those studied by Bakker ('59), and Nigon and Daillie ('58). There appears to be considerable variation in the time to pupation at 25°C reported by different workers — 90 hours (Bakker, '59), 96 hours (Bodenstein, '50), 110 hours L'Heritier and Teissier ('33), 104 hours for the Pacific strain and 112 hours for the strain studied by Nigon and Daillie. Differences in nutritional conditions could be partly responsible for such differences. It is also known that a high level of genetic variation exists in wild populations and that selection can lead to substantial differences in the duration of development, growth rate, body size etc. (Robertson, '55, '63). Biochemical analysis of a num-

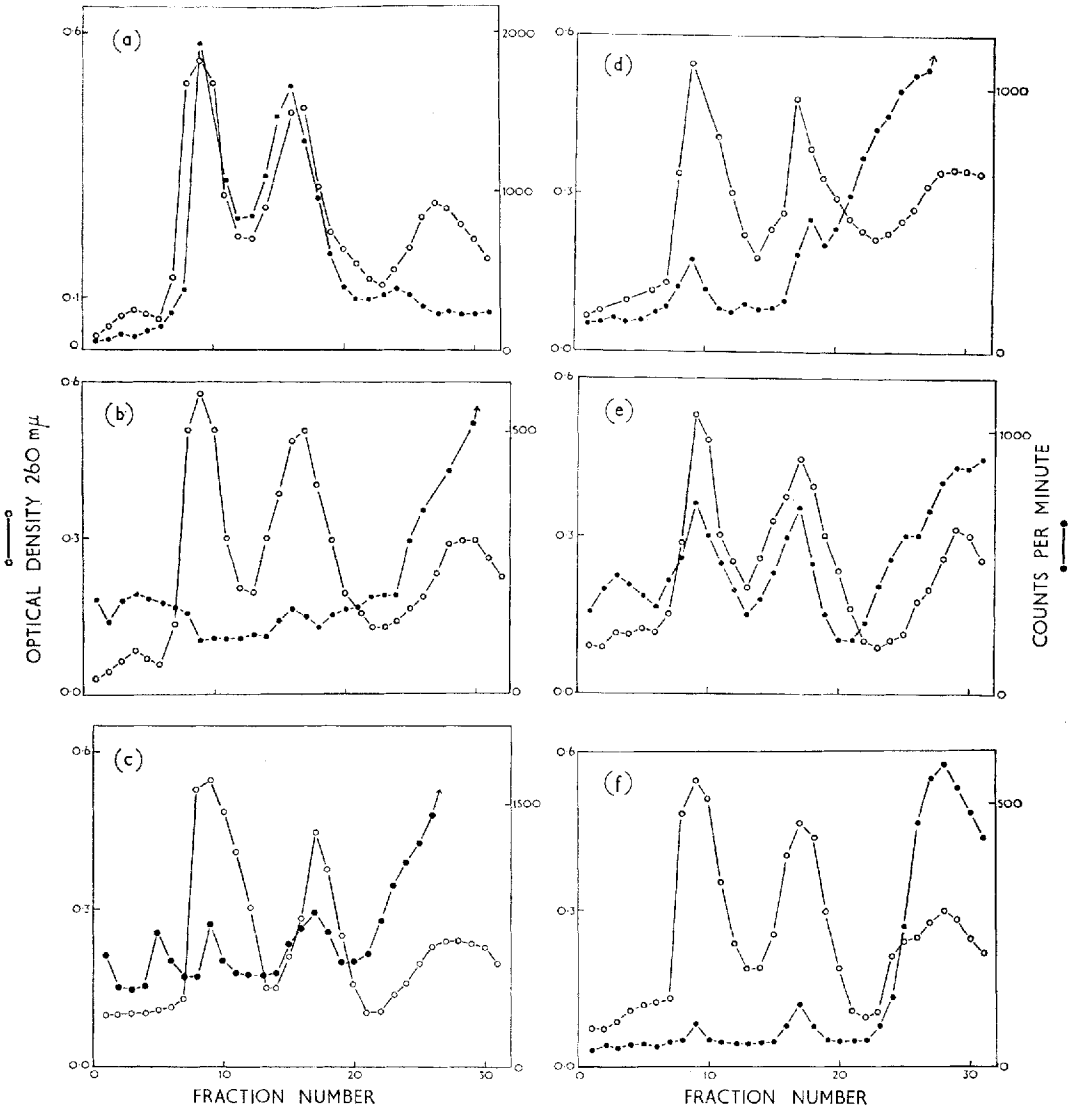

Fig. 9 The sucrose gradient, profiles of pulse-labeled (P^{32}) RNA, at successive stages of larval life, compared with *Drosophila* ribosomal RNA. —●— counts per minute, —○— optical density at 260 mμ (a) newly laid eggs, the marker in this run is provided by labeled mammalian ribosomal RNA; (b) newly hatched larvae; (c) 12 hour larvae; (d) 48 hour larvae; (e) 65 hour larvae; (f) 84 hour larvae.

ber of lines has shown that selection under appropriate nutritional conditions can lead to considerable differences in the DNA content per individual and in the protein/DNA ratio and also that such differences are related to characteristic differences between the lines in the duration of the larval period. (Church and Robertson, in press).

The evidence from *Drosophila* may be compared with similar growth data from other insect species. The protein content of insects, such as *Tenebrio* (Teissier, '31), *Aëdes* (Lang, Lau and Jefferson, '65) and *Calliphora* (Price, '65) is similar to that of *Drosophila* ranging from an initial 4% to a maximum of 10%. Many authors have reported differences in the

interrelations of the different nitrogen fractions, especially with respect to the complex protein and the free-amino acids in such species as the Japanese beetle (Anderson, '48), *Calliphora* (Evans, '32), and *Tenebrio* (Teissier, '31). Changes in the nitrogen distribution during development chiefly reflect the synthesis of adult protein. When differences in absolute development time and the number of instars are allowed for, the changes in *Drosophila* appear to be rather similar to those reported for other insects. The free amino content is high in the young larva, falls to a minimum at the start of pupation and rises again to eclosion. It is inferred that the reduction in the free amino-acids in the late larval stage is correlated with the growth of the imaginal discs. The increase during the pupal period may be essentially due to histolysis of larval tissues, but it must be noted that there is also a rapid increase in the thoracic musculature which accounts for over two thirds of the total protein in a newly emerged adult. (Miller, '50).

The increase in relative fat content during growth cannot be regarded simply as a measure of energy storage, since both phospholipid and neutral fats are represented (Finkel, '48), and only the latter can be regarded as stored energy. The total lipid fraction represents 15% of the wet weight of later larvae of *Drosophila* and this is in general agreement with the estimates reported for the meal worm (Finkel, '48), the tent caterpillar (Rudolphs, '26), *Tenebrio* (Teissier, '31), but is less than that reported for *Bombyx* by Uvarov ('28). Lang ('63) reported that the lipid content was increased when female mosquitos were cultured at a low temperature and that this may represent an adaptive change before hibernation.

The nucleic acid content of an organ is closely associated with basic metabolic differences in growth. When cells are diploid, the constancy of DNA per cell (Vendrely, '55) provides a useful standard reference in biochemical studies (Davidson and Leslie, '50), since the concentration of cell constituents may be related to the DNA content. In favorable situations, growth may be described in terms of cell size and number, since protein/DNA can be taken as a measure of cell size. The relative RNA content reflects cellular capacity for synthesising protein. The situation is complicated in holometabolous *Diptera* since most larval tissues grow by increase in the size of cells, which become polytene, whereas most adult tissue is diploid (Bodenstein, '50). It is an open question whether the protein/DNA ratio is influenced by whether or not replicated DNA is distributed between diploid cells or retained in a polytene nucleus. In the larval growth of *Drosophila*, there is a progressive fall in the protein/DNA ratio from about 60 hours to about 80 hours, when proliferation of imaginal discs is accelerated. Thereafter the ratio remains more or less constant to the adult stage. Since the cells of the larval tissues are increasing in size over this period, this might suggest a possible inverse relation between increasing polyteny and the protein/DNA ratio, but on the other hand, ontogenetic changes in the larval tissues may be the principal cause of this difference.

Since the protein/DNA ratio is stable after 80 hours, it appears that differences in body size between adults might be tentatively interpreted in terms of cell size and number from estimates of DNA and protein/DNA ratios.

Figure 10 represents an attempt to compare the DNA content during the development of the mosquito (Lang, Lau and Jefferson, '65), *Calliphora* (Price, '65),

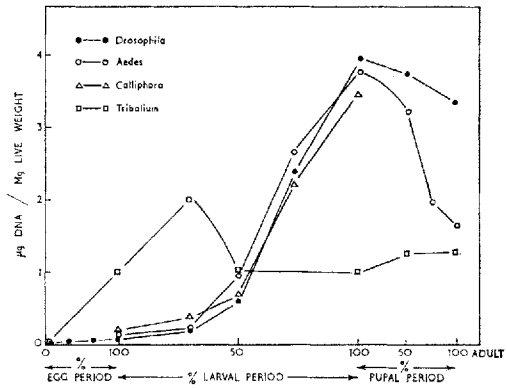

Fig. 10 Comparison of the ratio of DNA (microgram) to live weight (milligram) in different insects. The origin of the figures is described in the text.

Tribolium (Devi, Lamonde, Srivastava and Sakar, '63), with that of *Drosophila*. The data, except for *Drosophila*, have been estimated from the reports of the various authors, and, to allow for chronological differences, μg DNA per mg. wet weight is plotted against the percentage time for each developmental stage. The DNA estimates are only approximate since they were taken from graphs, and, in some cases, the weights of the animals referred to were not quoted so independent estimates of weight had to be used, while, in one case (Price, '65), nucleic acid phosphorus had to be converted to nucleic acid content without the information needed for an accurate conversion. Nevertheless, the estimates appear to be good enough to indicate major features.

The general pattern of DNA synthesis is common to all the *Diptera* examined, but *Tribolium* is apparently different, due either to substantial developmental differences or very early proliferation of imaginal tissue. The relative DNA content of adult *Drosophila* appears to exceed that of the other insects, but this difference may reflect inadequacy of the transformation of the data to a common scale and also differences in the method of estimating DNA. Brawerman and Shapiro, ('62) have reported that the DNA content of a diploid *Drosophila* cell is 0.17 picogram. Taking the average DNA content of an adult female as 1.013 μg and assuming the cells in adult terms are diploid, we can estimate the total number of cells as about 6×10^6.

The DNA content of the Pacific adults is less than that reported by Nigon and Daillie ('58). The difference may be accounted for by the axenic culture of the Pacific strain, which eliminates symbionts and other organisms. It is however surprising that no comparable difference has been detected for the corresponding estimates of RNA and protein. This apparent difference in DNA content, according to the cultural conditions, is receiving further study.

The RNA content of cells and tissues varies greatly in relation to physiological state (Leslie, '55; Brachet, '62). Brawerman and Shapiro ('62) found it difficult to arrive at a meaningful comparison of the RNA content in cells of different species, on account of the wide variation in the estimates reported from different laboratories. Their review shows that the RNA/DNA ratio varies in homologous tissues from closely related species but the pattern of a high ratio, in tissues which are synthesizing protein, and a low ratio, in rapidly proliferating tissue is quite general. Yao ('49) reported that organs characterized by polyteny and growth by increase in cell size in *Drosophila* are rich in RNA. At eclosion muscle cells contain little RNA although they have a high RNA content during metamorphosis.

A comparison of the RNA content in different forms, using the same kind of procedure as for DNA, is displayed in figure 11. All the species follow the same general trend in relative RNA content, rising rapidly to a maximum at a time which corresponds to completion of 50–75% of the larval period. Unlike the estimates for DNA, *Tribolium* is not atypical. Apparently *Drosophila* constitutes an exception since there is no evidence of a sharp decline in RNA/mg. wet weight from the late larval to the adult stage.

Thus, although only a few insects have been studied sufficiently thoroughly, all the *Diptera* follow a similar trend for either of the nucleic acids during development. Only the beetle *Tribolium* differs in the DNA/weight ratio. There is need for a wider comparative survey of biochemical composition during development in species

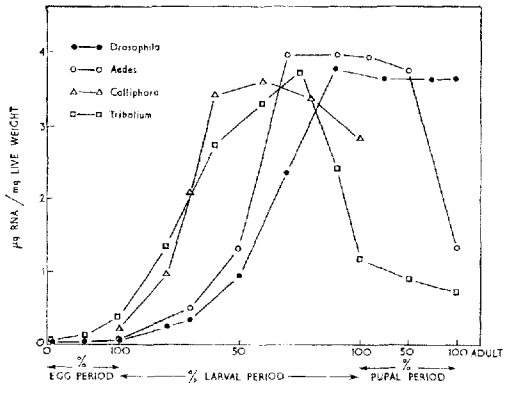

Fig. 11 Comparison of the ratio of RNA (microgram) to live weight (milligram) in different insects. The origin of the figures is described in the text.

from other insect families, including those with heterometabolous development.

The sucrose gradient analysis of RNA during development showed the usual 28 S, 18 S and also a 4 S RNA component although the quantity of the latter was obscured by various degradation products. Evidence for high molecular weight RNA is greatest in newly emerged larvae and also at 64 hours — both times when the rate of RNA synthesis is particularly high. To avoid possible contamination of labeled RNA by RNA from yeast or bacterial sources, it is essential to work with animals grown under axenic conditions. In this connection Brown and Littna ('64) noted that fecal bacteria, although numerically insignificant, nevertheless carried a high concentration of labeled RNA in *Xenopus*. The suggestive evidence for sequential changes in the relative concentration of rapidly labeled RNA paves the way for more detailed analysis. A holometabolous insect, such as *Drosophila*, provides specially interesting material for the study of nucleic acid synthesis during development, in view of the contrasts between the growth of larval and imaginal tissues.

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